MANGALORE REFINERY AND PETROCHEMICALS LIMITED

EQUIPMENT INSPECTION REPORT

Report No : INSP-2026-0447
Equipment : Crude Distillation Unit - Overhead Condenser E-101
Location : CDU Section, Refinery Block B
Inspection Date: 14 August 2026
Inspector : R. Shetty, Senior Inspection Engineer
Classification : CONFIDENTIAL - INTERNAL

1. SUMMARY OF FINDINGS

Visual and thickness survey of condenser E-101 shell and tube bundle was carried out during the scheduled shutdown.

FINDING F-1: Shell wall thickness measured 8.2 mm against the minimum required thickness of 8.8 mm. Localised thinning observed near the shell inlet nozzle. This is BELOW the minimum allowable thickness.

FINDING F-2: Tube bundle leakage detected at 4 locations during the hydro test at 12.5 bar. Pressure drop observed from 12.5 bar to 9.8 bar within 30 minutes. This exceeds the acceptance criterion of less than 5 percent pressure loss per hour.

FINDING F-3: Inlet nozzle gasket seat corrosion, approximately 15 percent circumferential coverage. Gasket replacement is required.

FINDING F-4: Protective coating integrity on the exterior shell is satisfactory. No action required.

2. RECOMMENDATIONS

R-1: Re-rate or replace shell section near inlet nozzle before the next operating cycle. Priority: HIGH
R-2: Plug or replace the 4 leaking tubes. Priority: HIGH
R-3: Machine and re-seat the inlet nozzle gasket face. Priority: MEDIUM
R-4: Continue routine external coating inspection. Priority: LOW

3. REFERENCES

Inspection carried out per Equipment Inspection SOP EMP-SOP-014 and the Pressure Vessel Integrity Manual PV-MAN-02.

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4. THICKNESS MEASUREMENT LOG (extract)

Location	Reading	Min req	Status
Shell circum.1	8.2 mm	8.8 mm	UNACCEPTABLE
Shell circum.2	9.1 mm	8.8 mm	Acceptable
Shell circum.3	9.0 mm	8.8 mm	Acceptable
Head north	9.4 mm	8.8 mm	Acceptable
Head south	9.2 mm	8.8 mm	Acceptable

5. HYDRO TEST RECORD

Test medium: water. Test pressure: 12.5 bar. Hold time: 60 minutes. Observed pressure after 30 minutes: 9.8 bar. Verdict: LEAK DETECTED.

6. PHOTOGRAPHS

Six photographs attached to the physical dossier (not included in this scan).

7. INSPECTOR DECLARATION

The findings above are a true record of the inspection performed on 14 August 2026.

Signature: R. Shetty Date: 14 Aug 2026